

# Grading of Prostate Whole-slide Images Using Weak Self-supervised Learning

1<sup>st</sup> Amirata Ghorbani  
Department of Electrical Engineering  
Stanford University  
Stanford, CA, USA  
amiratag@stanford.edu

2<sup>nd</sup> Andre Esteva  
Salesforce Research  
Salesforce  
Palo Alto, CA, USA  
aesteva@salesforce.com

3<sup>rd</sup> James Zou  
Department of Biomedical Data Science  
Stanford University  
Stanford, CA, USA  
jamesz@stanford.edu

**Abstract**—Prostate cancer is the second most common cancer among men worldwide. For prognosis and treatment, pathologists assign International Society of Urological Pathology (ISUP) grades to Whole Slide Imaging (WSI) of sampled prostate tissues to express the severity of the cancer. The manual approach suffers from human error and person-to-person variations. Existing approaches that use machine learning for automatic grading of WSI images includes using expensive datasets in which different regions of the slide are annotated by pathologists to show different levels of cancer. However, most of the existing real-world datasets contain weak labels; i.e. each slide is labeled with just one grade number. In this work, we present a self-supervised learning method to automatically grade prostate slides using datasets of weakly labeled slides. Our algorithm achieves a 5-way classification accuracy of 54.1% which surpasses the performance of 41.5% using state of the art multiple-instance labeling methods.

**Index Terms**—ISUP grading, Weak Supervision, Self-supervised learning, Representation Learning

## I. INTRODUCTION

Prostate cancer is the second most common cause of cancer death in American men. For a patient suspected to have prostate cancer, the prognosis and treatment depends on a number of data points. The most important data points is the ISUP grade which is assigned by a pathologist by examining needle biopsy samples of prostate [1]. To compute the ISUP grade, an abnormality score (called Gleason score) is assigned to different cell regions in the biopsy slide. Given the abnormality core of two most common patterns in the slide, the 5-level ISUP grade is calculated. ISUP grade 1 is the lowest risk score and ISUP grade 5 is the highest.

Similar to most manual tasks in medicine, ISUP grading process can suffer from human error and pathologist-to-pathologist variation [2]–[5]. Automating the ISUP grading process using Artificial Intelligence (AI) can mitigate the issues of error and variation [6]–[9]. In order to automate the ISUP grading using supervised machine learning, the intuitive but expensive approach is to train the ML model to mimic the grading procedure of pathologists. That is, the model should assign an abnormality score to each cell region in the slide and then aggregate the scores to output the ISUP grade [7], [10], [11]. The issue with such approach is that to train the ML model, one needs to have a large dataset of slides where all cell regions of the slide are annotated with high

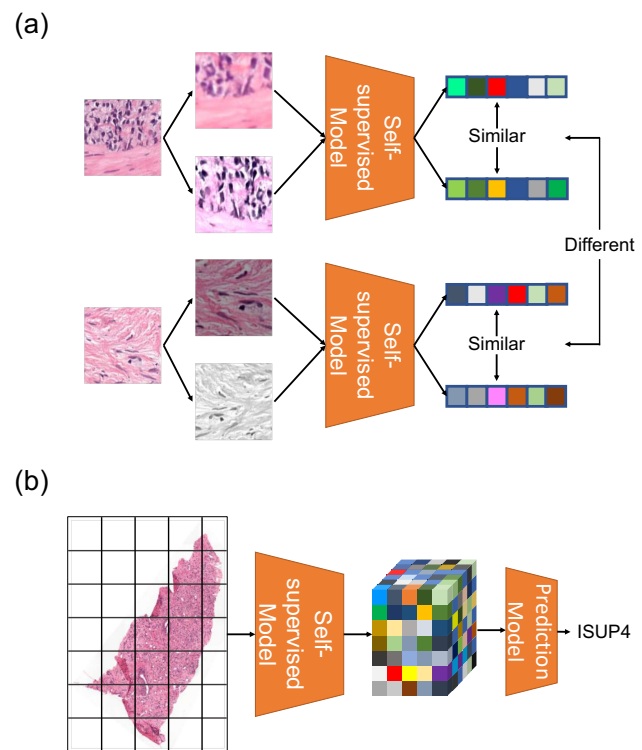


Fig. 1. **Overview of our two-stage model.** (a) At first, all training Slides are divided into patches to train the first-stage model. At each iteration of training, a batch of patches is selected randomly. Each patch is transformed twice with a random set of image transformations (e.g. flipping, rotation, contrast change, etc) to create two augmented versions of the same image. The batch of augmented images is mapped to a batch of embedding vectors. The contrastive loss function is such that the embedding of images that are originated from the same patch should have a smaller cosine distance compared to all other patches. (b) After learning the SSL model, the prediction model is trained by mapping patches of each slide to their respective embeddings. The results is a three-dimensional tensor that preserves the original slides' spatial information. At the last stage, a prediction model is trained on the embedding tensors to predict the label of the whole slide.

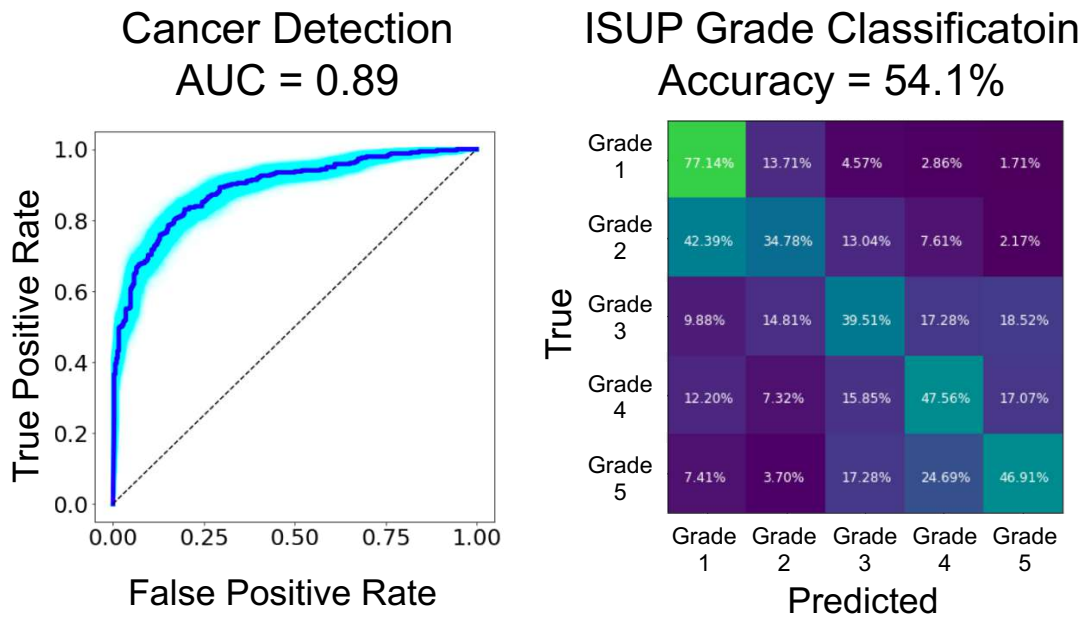


Fig. 2. **Prediction results.** After training the first-stage SLL model, we train two prediction models for two different tasks. The first task is the binary classification of whether a slide is cancerous or not. The left figure shows the ROC curve. As our second task, we train a second-stage model that classifies the cancerous slides into the 5 ISUP grades. The right figure shows the confusion matrix of the model on the test set.

resolution abnormality scores. Creating a such rich dataset is very expensive as pathologists must go over each region in each slide and annotate it. Most of the existing electronic health record (EHR) data, however is weakly labeled; that is, instead of having high resolution annotations of all cell regions in the slide, there is only a single number (ISUP grade) assigned to each slide. In order to be able to utilize the large amount of available EHR data, we need to develop weakly-supervised ML systems that can predict the ISUP grades by using weakly labeled training data.

## II. OUR CONTRIBUTION

We introduce a two-stage self-supervised learning algorithm for weakly supervised learning of ISUP grades of pathology whole slide images. We show that our algorithm is able to classify slides into ISUP grades using only slide-level labels. The first stage of our algorithm is a self-supervised contrastive learning model trained on small patches extracted from pathology slides. The second stage takes the slide transformed by the first model and assigns an ISUP grade to the slide. We then examine the self-supervised learning model further and show that it learns meaningful features without any explicit annotation of the slides. We also show that our algorithm is interpretable and by using existing saliency map methods we demonstrate that the model's decision-making is based on related information in the slide.

## III. RELATED LITERATURE

Recent literature has successfully implemented deep learning for automatic detection of malignant tissue in various pathology tasks [12]–[14] and in wide range of tissue types

like breast [15], esophagus [16], thyroid [17]. Grading of prostate pathology slides using deep learning has been studied. Earlier works [18] focused on the easier task of grading homogeneous slides or binary classification of high vs low Gleason score slides. Most of the existing work is based on first training a model on individual patches that annotated with a Gleason score and then aggregating the individual patch predictions into a WSI label. [6], [7]. These approaches require pathologists to provide high-resolution region-level annotations of Gleason scores. Following works have focused on making the annotation process cheaper by the use of automatic annotation approaches. [8] A recent effort has shown the benefits of using automatic ISUP grading systems to assist pathologists. [9]

Our work is similar to the body of work that does not rely on expensive region-level annotations. One approach is to extract manual features [19]. A more recent approach [20] is to use multiple instance labelling (MIL) where the idea is train two models simultaneously. The biggest issue with the MIL approach is that it erases the spatial relationship of different regions in the slide, which we preserve. Our approach is also different from the existing methods in that we use the recent powerful advances of self-supervised learning [21]–[23].

## IV. METHOD

Our method is a general approach that works for weakly supervised learning of high resolution images (in this paper, we are focused on prostate pathology slides). The overall idea is to compress the visual information in the the image while preserving its spatial structure. Our algorithm has two stages: the self-supervised learning stage and the prediction stage.

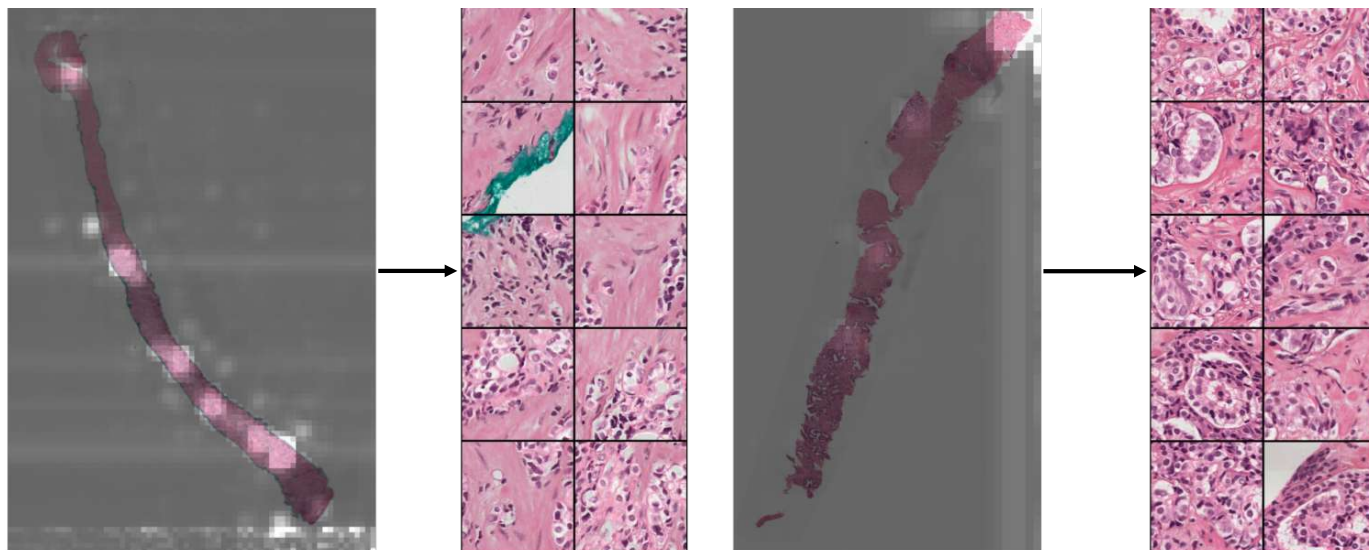


Fig. 3. **Interpretability examples** We have shown two examples of slides with the highest ISUP grade (group 5). For each slide, we apply the SmoothGrad interpretability algorithm to the second-stage model and find the patches with the highest importance scores. It can be seen that as expected, the important regions are a few clusters of patches across the slide which is exactly similar to the way ISUP grades are computed by pathologists. The highlighted regions contain patches that have the highest importance scores. We randomly select 10 patches from the top 1% of highest scored ones and show them here. For both slides, all of the depicted patches contain cancerous cells.

#### A. Machine Learning for High Resolution Images

In image classification, each image is assigned to a label and the ML task is to train a model to predict the label given the image (e.g. Imagenet [24]). For classification of high resolution images (e.g. pathology slides), due to computational constraints, it is not possible to feed the whole image to a deep learning model. However, as for pathology slides, the local granular information of individual patches of cells is important for diagnosis. Therefore, lowering the image resolution will result in degraded performance and will not solve the issue. The best performing existing approach is based on multiple instance learning (MIL). The idea is to break down the slide into smaller *patches* and assigning the slide label to this *set* of patches [25]. The MIL model then uses an attention model on top of an embedding extractor model to select the top predictive patches. The embedding of the top patches is then averaged and used by a fully-connected layer to output a single label for the slide's set of patches. Both models are trained simultaneously in an end to end fashion. Although this approach has performed well in some pathology tasks [13], it does not perform well for ISUP grade prediction. The reason is that the MIL approach removes the spatial structure of the slide which is necessary considering how ISUP grades are assigned by pathologists. To solve this problem, we introduce a two stage SSL-based model. The first stage of the model learns meaningful low-dimensional (in our experiments, 128 dimensions) *embeddings* of patches. Therefore, by replacing each patch with its embedding, the slide is compressed while the information in each patch and the spatial relationship between patches is preserved. We then train a second-stage model to predict the slide's label from the 3-d tensor of stacked embeddings. This model does not suffer from computational

limitations as the 3-d tensor is much smaller (around 400 times) than the original slide. Fig. 1 shows an overview.

#### B. Self-supervised learning model

The idea of self-supervised learning is to learn an informative embeddings from unlabeled images. We use a ResNet50 model [26] as the backbone of our SSL model. Each slide is divided into  $224 \times 224$  patches resulting in an unlabeled dataset of millions of patches (Fig. 1(a)). During training of the SSL model, each patch is transformed twice with a two sets of random image transformations like adding noise, rotation, contrast change, etc. The result is two images that look very different but contain the same information. The model is trained to map the two augmented versions of the same image to similar embedding vectors. We use the MOCO algorithm discussed in [22].

#### C. Prediction model

The patches of each slide are replaced by the SSL model's extracted embedding. For instance, if the original slide is  $2240 \times 2240 \times 3$  dimensional, it will be compressed into a tensor of size  $10 \times 10 \times 128$  (Fig. 1(b)). The prediction model is then trained in a supervised fashion to predict each compressed slide's ISUP grade. We use a deep learning model with three convolution layers with a max-pooling layer after each convolution.

### V. EXPERIMENTS AND RESULTS

#### A. Dataset

In this work, we use the famous Prostate cANcer grade Assessment (PANDA) Challenge dataset<sup>1</sup>. The comes from

<sup>1</sup><https://www.kaggle.com/c/prostate-cancer-grade-assessment>

two different sources: Radboud University Medical Center with 5160 prostate slides and Karolinska Institute with 5465 slides. Each slide is labeled with an ISUP grade between 0 and 5. An interesting point is that there are slight differences in the scanned slides as the staining and scanners between two institutes is different. Slides are high resolution with variations in size (thousands of pixels by thousands of pixels) which means that it is not possible to feed the slides directly into the ML model.

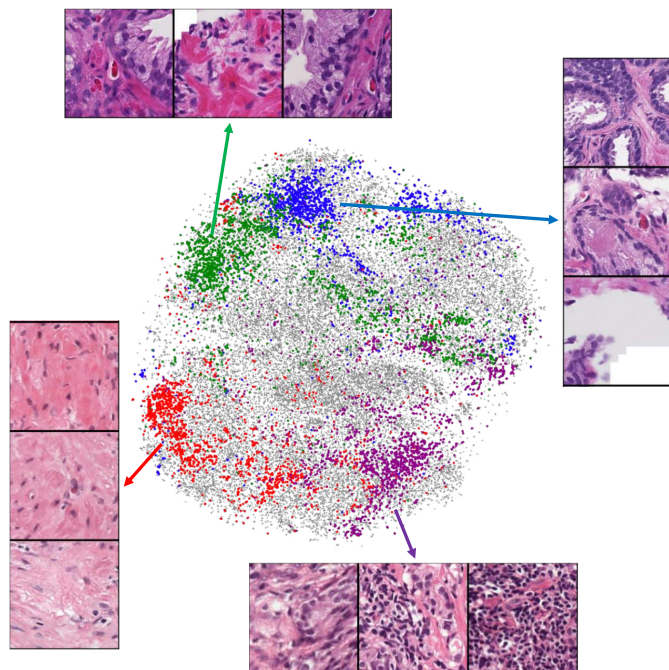


Fig. 4. **SSL model visualization** In order to examine the learned representation of the SSL model, we randomly sample 40000 patches and extract their SSL representation. The figure shows the t-SNE [27] transform of the representations. To find the clusters of patches that the SSL model represents similarly, we apply K-means clustering to the representation vectors. We examine four of these clusters by showing three randomly sampled patches. The blue and green clusters consist of tumoral nucleus and lymphocytes while the red cluster consists of stromal regions.

### B. Prediction performance

We train the two stage model for classifying slides into five ISUP grades. The model achieves an overall accuracy of 54.1% on this task. Fig. 2 shows the confusion matrix of the model. As expected, model's mistakes mostly happen between similar grades. We also train the second stage model for the binary task of classifying healthy slides from ones with any grade of prostate cancer. This binary classification model achieves an AUC score of 0.89.

To show the importance of the SSL step, we use the same two-stage structure but instead of training the first model by SSL, we use the weights of a model that is pretrained on ImageNet [24] dataset to extract patch embeddings. This approach achieves an ISUP classification accuracy of 41.4% which is significantly lower than the SSL approach.

To compare our approach with the MIL approach, we implement the model discussed in [13]. This approach trains the two models in an end-to-end fashion without the use of SSL. The first model is trained to extract patch embeddings while the second model is an Attention model. This model achieves an accuracy of 41.5%.

### C. Interpretability

Another way to examine our two-stage model is to interpret the model's decision making. The idea is to find the areas that are most salient for the model's decision-making. To this end, we apply the popular SmoothGrad [28] interpretability method to the second-stage model. This method assigns an importance score to each input representation vector (and as a result its corresponding patch). Fig. 3 shows two slides with ISUP grade 5 where the saliency map of patches is overlaid on the slide to show the salient regions. We also show 10 examples of highly important patches which are selected randomly from top 1% of patches with highest saliency score. All of the patches contain cancerous cell regions.

### D. Examining the SSL model's embeddings

The idea behind the SSL model is that through the use of contrastive loss, the model is able to map similar patches of cells to similar embedding vectors. As a subjective examination, we select a random set of 40000 patches from our training dataset and map them through the SSL model. We then apply K-means clustering to the embedding vectors and examine each cluster. Fig. 4 shows the TSNE visualization of these patches and four example clusters. Further analysis shows that these clusters contain coherent and meaningful set of patches with similar types of tissue.

## VI. DISCUSSION AND FUTURE WORK

In this work we introduced a self-supervised learning model to classify high resolution images of prostate pathology slides into different levels of prostate cancer severity. We showed that its performance is superior to the existing MIL-based benchmark as it preserves the pathology slides' spatial information. A possible next step is to apply this approach to other pathology tasks in different tissue types. Another next step would be to develop self-supervised learning models that are specifically designed for the specific type of data that pathology slides contain; models that are trained using image augmentations that work best for pathology patches.

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